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Metagenome metabolic analysis revealing the mechanism of simultaneous methanogenesis, aerobic methane oxidation and denitrification (SMAMOD) in a microaerobic up-flow sludge bed biofilm reactor

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Abstract

BACKGROUND: Improved denitrification with methane as alternative electron donor is regarded as a promising low-cost approach for organic nitrate-containing wastewater treatment. However, *in situ* capture of methane released from a methanogenic system for enhancing nitrogen removal seems to be scarcely reported. The applications of simultaneous denitrification, methanogenesis, and aerobic methane oxidation coupled to denitrification provide the possibility for solving the challenging issue.

RESULTS: In this study, simultaneous methanogenesis, aerobic methane oxidation, and denitrification (SMAMOD) was firstly established in a microaerobic up-flow sludge bed biofilm reactor (MUSBBR) treating high-strength synthetic wastewater with 2000 mg L^{-1} Chemical oxygen demand (COD) and 400 mg L^{-1} nitrate. The operation demonstrated 98.8% COD, 98.8% NO_3^--N, and 93.5% TN removals were stably obtained. Microbial community found the coexistence and cooperation of *Methanothrix* as methanogens, *Methylogaea* as aerobic methanotrophs, as well as facultative denitrification genera in SMAMOD system. Gene profiling demonstrated key functional genes involved in aerobic and anaerobic fermentation, methanogenesis and aerobic methane oxidation, as well as nitrate reduction induced by nitrate reductase systems. In organic oxygen-limited biotransformation, glucose was degraded mainly through aerobic respiration, anaerobic fermentation into methane by decarboxylation of acetic acid, and further methane aerobic oxidation pathways, while acetate- and methanol-denitrification and dissimilatory nitrate reduction to ammonium pathways participated in the nitrogen bioconversion. Further, the metabolic mechanism of SMAMOD might be unraveled by spatial interactions of key metagenomes in core microbial consortiums in the MUSBBR reactor.

CONCLUSION: The study offers valuable metagenomic insights into SMAMOD as a potentially compact process for sustainable treatment of high-strength organic nitrate-containing wastewater.
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Keywords: SMAMOD; aerobic methane oxidation; metagenomics; functional genes; metabolic mechanism

INTRODUCTION

A large amount of nitrate (over 200 mg NO_3^--N/L) is generally contained in complex high-strength organic wastewaters produced from the dairy, explosive, fertilizer, and nuclear industries, as well as landfill industries,[1] so nitrogen removal needs to be considered during conventional anaerobic treatment of organic matter.[2] In tradition, a separate denitrification process is often indispensable after anaerobic digestion (AD) treatment of these wastewaters.[3] However, this practice seems not economic and sustainable. For one thing, the supply of external carbons is needed for denitrification, because most of the biodegradable COD is removed beforehand by anaerobic treatment, thus increasing the subsequent removal of residual organics and operational costs. Another point is that some of this methane, especially dissolved methane which is difficult to recycle, may lead to powerful greenhouse gas (GHG) effects if improperly managed.[4] Therefore, it is essential to eliminate nitrogen and dissolved methane from organic- and nitrate-containing wastewater during AD treatment.

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Simultaneous denitrification and methanogenesis (SDM) was a feasible alternative to post-denitrification through introducing denitrification into the methanogenic reactor.[5] In the single SDM reactor, a fraction of COD is able to be utilized as electron donor for nitrate reduction without installing an additional denitrification unit. Until now, SDM has been achieved in anaerobic baffled reactor, [6] anaerobic up-flow filter,[7] up-flow anaerobic sludge blanket,[8] and expanded granular sludge bed.[9] However, the competition of organic matter between denitrifiers and methanogens largely leads to affect SDM performance.[5,8] Over the past decades, methane oxidation as a newly global carbon pathway has been extensively discovered in natural water environments, [10] which plays a significant role in methane emission reduction. Under anaerobic or aerobic environments, diverse groups of methyl-oxidizing bacteria (methanotrophs) allow methane oxidation.[11] In contrast to anaerobic methanotrophic bacteria (AnMB), aerobic methanotrophic bacteria (AMB) demonstrate intrinsic advantages with faster growth rate, stronger metabolic ability, and less environment sensibility.[12] Aerobic methane oxidation (AMO) means that methane is enabled to be oxidized into some organic intermediates (such as methanol), which can supply alternative organic carbons for denitrification, known as aerobic methane oxidation coupled to denitrification (AME-D).[13] So far, AME-D process has been investigated in some bioreactors with external methane gas supplied by either pumping aeration[14] or membrane aeration[15] into the liquid.

If some portions of excessive methane difficult to collect and reuse are capable of being utilized by AMB, it is more likely to improve nitrogen removal capacity and mitigate GHG potential by integrating AMO and denitrification into a single methanogenic reactor. Hence, from the viewpoint of availability and economy, *in situ* utilization of methane generated in methanation systems will be undoubtedly more attractive to the application of AME-D. Due to the low solubility of methane in the liquid phase,[16] however, high methanogenesis efficiency should be maintained to guarantee enough capture of methane for possible *in-site* AMO. Previous literatures[17,18] have reported the effectiveness of microaerobic pretreatment for improving methane yield, since facultative *in-situ* can rapidly hydrolyze organic matter into small molecule organics, such as volatile fatty acids (VFAs), in favor of the growth of methane-producing bacteria (MPB). Moreover, microaerophilic maintenance also serves as a pivotal link between AMO and denitrification for achieving AME-D. Based on these, methanogenesis, methane aerobic oxidation, and denitrification are expected to simultaneously (SMAMOD) occur in one microaerobic system.

As an effective tool to deeply understand the fundamentals of complex wastewater biological treatment processes, 16S rDNA-based metagenomics were used for identifying interactions of the mixed microbiomes and determining key functional gene metabolic pathways driving biochemical reactions.[19,20] Therefore, the present study mainly aims to reveal the underlying metabolic mechanism of SMAMOD using metagenomics based on the successful build-up of SMAMOD in a single microaerobic up-flow sludge bed biofilm reactor (MUSBBR).

MATERIAL AND METHODS

Reactor start-up and operation

A lab-scale MUSBBR system was used in Fig. 1. The 17 L effective volume system consisted of the main reaction body (9 L) and a three-phase separator (8 L). The main body included a sludge bed combined with a packing bed filled with about 25% (v/v) PU sponge as biocarrier. The seeding sludge was obtained from a municipal wastewater treatment plant (Jingzhong city, Shanxi province) and inoculated into the reactor at 2:1 mixing ratio of thickening sludge (MLSS: 15.76 g L^{-1}) and aerobic activated sludge (MLSS: 7.47 g L^{-1}). In this study, synthetic wastewater contains tap water and specific compositions (L) as follows: 2134 mg glucose, 0–6071.4 mg $NaNO_3$, 50 mg $NaHCO_3$, 309.7 mg NaH_2PO_4, 55.5 mg $CaCl_2$ and trace element solution (5 g EDTA, 2 g $FeCl_3{\cdot}4H_2O$, 0.2 g $CoCl_2{\cdot}6H_2O$, 0.5 g $MnCl_2{\cdot}4H_2O$, 0.03 mg $CuCl_2{\cdot}2H_2O$, 0.05 g $ZnCl_2$, 0.1 g Na_2SeO_3, 0.05 g $NiCl_2{\cdot}6H_2O$, 0.09 g Na_2MoO_4, 0.05 g H_3BO_4 per liter). To ensure mesophilic methane production, the wastewater temperature was controlled at 35°C by a heating controller and influent pH was adjusted to 7.2[21] by adding HCl (1 mM). During the start-up, the reactor was only fed with high-strength organic wastewater (2000 mg L^{-1} COD) without nitrate. After a 48-day stable methanation start-up, nitrate (0–1000 mg L^{-1} NO_3^--N) was gradually added into the feeding. According to a preliminary experiment, the optimal performance occurred at the corresponding influent COD/NO_3^--N (C/N) of 5. The pre-aeration before the feeding flowed into the reactor was undertaken to maintain microaerobic conditions. Dissolved oxygen (DO) and oxidation–reduction potential (ORP) levels were 0.2 mg L^{-1} and – 340 mv, respectively. The hydraulic retention time (HRT) was fixed at 2 days during the experiment period.

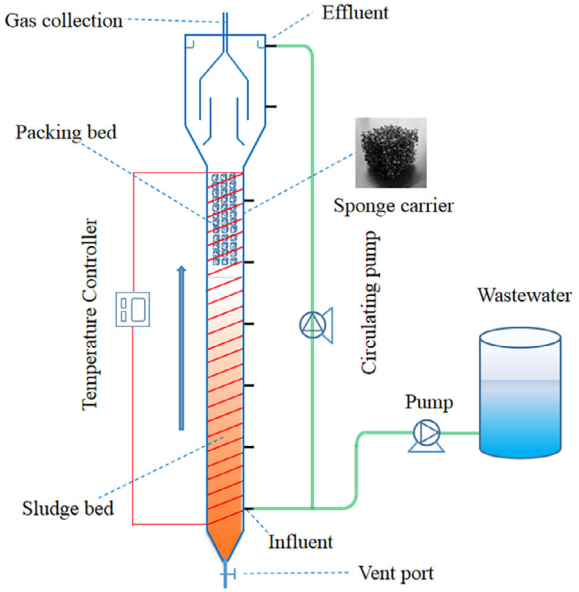


Figure 1. Schematic diagram of reactor system.

NH_4^+-N, NO_2^--N, NO_3^--N and COD of the effluent samples after 0.45 μm membrane filtering were determined by cuvette kits test (HACH LANGE LTD, US). Total nitrogen (TN) is the sum of NH_4^+-N and NO_x^--N. pH, DO, and ORP were monitored by a portable analysis meter (WTW 3420, Germany). CH_4 and CO_2 gas samples over the headspace were analyzed by a gas chromatograph (GC-4000A, China) equipped with flame ionization detector (FID) and thermal conductivity detector (TCD), respectively.

Sample collection and DNA extraction

Three different sludge samples with 10 mL each were collected from top, middle, and bottom sample ports from the sludge

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bed, and nine bio-carriers were collected from top, middle, and bottom sample ports from the packing bed on day 32. All the sludge samples were directly mixed and the mixture was put into a 2 mL centrifuge tube for DNA extraction, while all the carriers were cut and mixed. Biofilm surface microorganisms were ensured to be contacted using the lysate to release more DNA. Specific DNA extraction steps were according to OMEGA kit E.Z.N. Atm Mag-BindSoil DNA Kit's usage instructions (http://omegabiotek.com/store/production/soil-dna-kit/).

High-throughput sequencing

Polymerase chain reaction (PCR) amplification was carried out. 16S rDNA amplification area for V3-V4 domains, PCR primers for bacteria incorporating Miseq sequencing platform V3-V4 universal primers 341 F primer: CCCTACACGACGCTCTTCCGATCTGCC TACGGGNGGCWGCAG, 805 R primer: GACTGGAGTTCCTTGGC ACCCGAGAATTCCAGACTACHVGGGTATCTAATCC. The Qubit 3.0 DNA testing kit was used to precisely quantify the recovered DNA. When mixed with the same amount, 10 ng of DNA was taken from each sample, and Illumina Miseq high-throughput sequencing was performed by Sangon bioengineering co., LTD (Shanghai, China). The compiled sequences were compared with the rRNA gene sequences in the GenBank using NCBI BLAST program. The sequences were clustered into operational taxonomic units (OTUs) with the 97% similarity threshold.

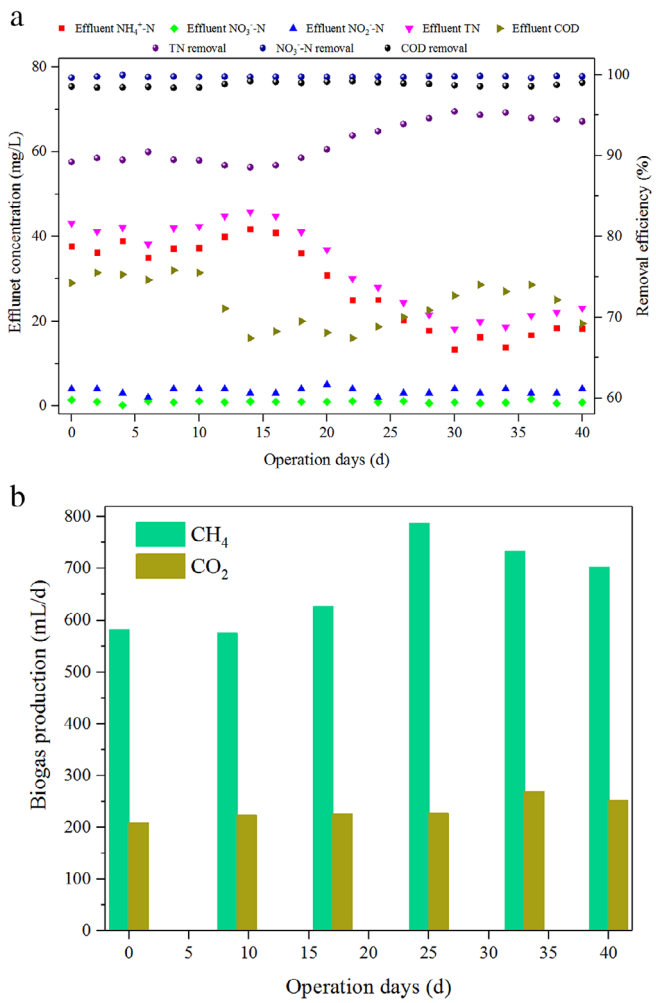


Figure 2. Reactor performance: a Effluent COD and nitrogen and removal efficiency; b Biogas production.

Bioinformatic analysis

The raw image data file obtained by Illumina Miseq™ was transformed into the original sequencing sequence by CASAVA Base Calling analysis and stored in FASTQ (fq) format file, containing the sequence information and its corresponding quality information. To pretreat the raw data, it was necessary to remove the primer joint sequence, then splice the paired reads into a sequence according to the overlapping relationship between the PE reads, then identify and distinguish the samples according to the barcode technical label sequence to obtain each sample data. Finally, the quality of each sample data was controlled and filtrated to obtain the valid data. Subsequently, the unamplified region sequence in the pre-processed sequence was removed using Usearch, followed by sequencing error correction of the sequence, and the uchime was used to identify the chimera. The chimeric removal sequences with the database representative sequence were compared by BLAST. To obtain the metabolic information, PICRUSt software was used to predict the functional gene composition in the samples obtained by 16 s sequencing to analyze the functional differences between different samples. The protein sequences of predicted genes were compared by Kyoto Encyclopedia of Genes and Genomes (KEGG) database (https://www.kegg.jp/).

To explore organic and nitrogen metabolisms, associated functional genes involved in metabolic pathways were identified in detail according to protein sequences predictions using KEGG database.[22] Key genes for fermentation chiefly include glucose kinase (glk), glucose phosphate isomerase (pgi), phosphate glycerate mutase (gpm), pyruvate formate lyase (pfl), pyruvate kinase (PK), pyruvate dehydrogenase (pdh), pyruvate ferredoxin oxidoreductase (por), and 2-oxoacid ferredoxin oxidoreductase subunit (oor). Key enzymes involved in the tricarboxylic acid (TCA) cycle are citrate synthase (CS), isocitrate dehydrogenase (IDH), 2-oxoglutarate ferredoxin oxidoreductase subunit (kor), and malate dehydrogenase (mdh). Key methane-producing and methane-oxidizing genes involve acetokinase (ack), phosphoric acetyltransferase (pta), acetyl-coa synthetase (acs), CODH/ACS complex subunit (cdh), tetrahydromethanopterin methyltransferase (mtr), methyl coenzyme M reductase (mcr), heterodisulfide reductase (hdr), F420 non-reducing hydrogenase (mvh), methane oxygenase (mmo), alcohol oxidase (MOX), formaldehyde dehydrogenase (fdh), formic acid dehydrogenase (FDH, fdw, fds) and glutathione dehydrogenase (frm). [23] In the nitrogen cycle, key enzymes include: nitrate reductase (nar, nap), nitrite reductase (nirKS), nitric oxide reductase (nor), nitrous oxide reductase (nos), nitrite reductase (nrf, nirBD), assimilatory nitrate reductase (nas, nirA), and nitrogenase (nif). [24] These genetic results were converged into KEGG metabolic networks to reveal metabolic pathways and their roles in carbon and nitrogen cycles.

RESULTS AND DISCUSSION

Reactor performance

Figure 2 shows the system performance under the influent 400 mg L^{-1} NO_3^--N. Under the steady-state condition, the system averagely removed as high as 98.8% COD with a high methane recovery up to 664.5 mL day^{-1}, indicating stably efficient methanation efficiency. Meanwhile, nearly 99% nitrate removal rate was also achieved. In the effluent, ammonia nitrogen was basically in

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the form of total nitrogen, while nitrate/nitrite nitrogen was rather low. As such, the corresponding average TN removal efficiency reached up to 93.5%. In theory, it was not possible to achieve high methane yield at such a low C/N ratio (<10), since most of the organics would be consumed for denitrification in SDM systems.[5] Yet this could not explain desirable methane production. Therefore, we inferred that in a robust methanogenic system, stepwise nitrate addition had a minor effect on methanogenic activity. Moreover, microaerobic conditions accelerated a large accumulation of organic acids and resulted in a sharp fall of the pH level (<7), thereby fermentation bacteria outgrew denitrifiers at low pH levels. These factors led to the precedence of methanogenesis over denitrification in our system. But if only methanogenesis occurs, excellent nitrogen removal seems impossible. Thus, we could conclude the significant role of methane in nitrogen removal promotion *via* aerobic methane oxidation coupled with denitrification (AME-D), when organic substrates severely lacked. Based on this assumption, most of the glucose should be fermentatively used for methane transformation instead of denitrification. Then, methane *in situ* generated in a methanation reactor was oxidized by aerobic methane-oxidizers into oxidative products, such as methanol for nitration reduction under oxygen-limited conditions. Besides, nitrate being directly reduced to ammonia *via* DNRA, frequently happened in SDM systems fed with glucose.[5,25] That could explain a bit of ammonia left in the final effluent. Given these factors, we forecast that methanogenesis, aerobic methane oxidation, and denitrification should be concurrently developed in a single methane-producing system through integrating SMD and AME-D.

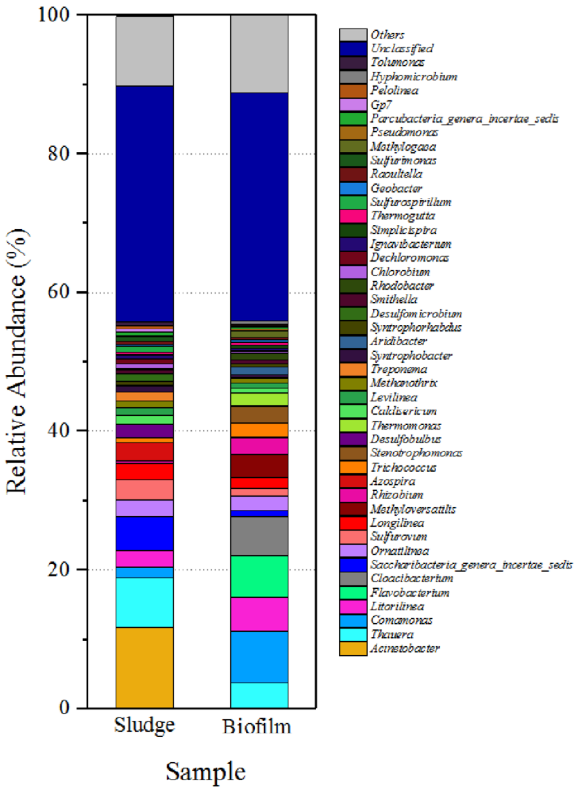


Figure 3. Microbial community compositions at genus level.

Microbial community analysis

To identify the functional bacteria in the SMAMOD system, the sludge and biofilm samples were respectively taken from the sludge and packing bed for microbial community analysis based on 16S rRNA gene sequencing. The count of OTUs was found to be 5450 for the sludge sample and 4779 for the biofilm sample. Also, the Shannon index of 5.61 was in sludge, while 5.14 in biofilm. These data imply that the sludge bed has more complex and diverse community structures than the packing bed. At phyla level, *Proteobacteria* were largest, followed by *Chloroflex* and *Bacteroidetes*. These major phyla suggest the dominance of facultative heterotrophs responsible for organic matter and nitrate transformations. Furthermore, *Euryarchaeota* were also discovered in either sludge or biofilm samples, certifying the presence of methanogenic microorganisms.[26] Figure 3 further displays microbial compositions at genus level. Among the total of 389 genera, some genera *Thauera*, *Comamonas*, *Rhizobium*, and *Rhodobacter* as members of chemoheterotrophs were shared in both samples. Remarkably, *Methanothrix*, as a common methanogen that thrives in acetate-rich environments[27] was discovered in both two samples, denoting widespread development of methanogenesis throughout the reactor. For the whole population structure, however, there was some evident differentiations between two samples. It was interesting to note that *Methylogaea*, belonging to aerobic methanotrophs[28,29] overwhelmingly existed in the biofilm (0.92%), whereas almost none existed in sludge. This exciting discovery verified the potential of AMO in the system. AMO might be encouraged by the prolonged contact time of methane and an enlarged low-oxygen micro-environment provided by a packing bed. Furthermore, *Methyloversatilis* (3.4%), as an alcoholic denitrifer[30] also typically found in biofilm, confirms methanol produced from AMO for denitrification *via* AME-D. Despite the richness of *Acinetobacter*, *Saccharibacteria* with strong degrading capacity of complex organic matter[31,32] were far higher in sludge than those in biofilm. This might be because aerobic and anaerobic glucose degradation initially occurred at the sludge bed of the micro-aerobic system, where the organic substrates were quite abundant. Other dominant facultative genera *Thaura*, *Comamonas*, *Flavobacterium*, *Longilinea*, *Azospira*, and *Thermomonas* were also extensively identified. These nitrogen-cycle-associated microbes were closely related to nitrogen removal. In general, the 16S rRNA gene sequencing consequences gave strong evidence for microbial co-existence-induced SMAMOD in a MUSBBR system.

Functional gene profiling

Further functional gene profiling during the organic matter and nitrate biocoversion in the system is illustrated as Fig. 4. As clearly shown, substantial fermentation genes and subunits were dominant in both samples, while more associated genes numbers were observed in the sludge. That was associated with the predominance of fermentative bacterium carrying these genes with the ability of degrading glucose. Particularly, a higher abundance of anaerobic and aerobic fermentation functional genes implied further anaerobic conversion of glycolysis products to methane as well as a few its aerobic degradation to CO_2 *via* TCA cycle. Moreover, the abundance of reductase subunit (hdr) in sludge was up to 29754, which was much higher than that in biofilm (15832). This indicated that methane production firstly took place in the sludge bed. Due to the presence of functional genes related to the conversion of acetic acid into methane, acetotrophic methanogenesis was proved in this study, in line with the observation of *Methanothrix*. In addition, vast key genes mmo with the potential function of methane oxidation[33] were significantly monitored

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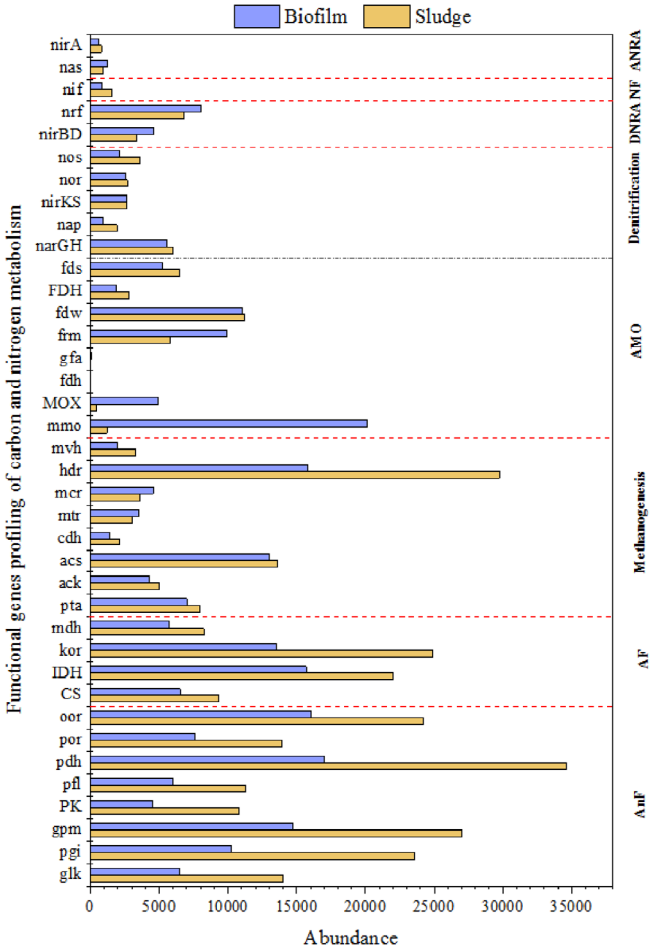


Figure 4. Functional genes profiling of carbon and nitrogen metabolisms.

in the biofilm. The whole detection of genes (mmo and MOX,[34] fdw and fds) also provides a complete evidence chain for the formation of AMO in the packing bed of the MUSBBR system.

Compared with carbon metabolism genes, nitrogen-transformation-associated functional genes were relatively low. Among them, a higher abundance of nitrate reductase (nar) with more than 5000 was found either in the sludge or biofilm, indicating a high nitrate removal rate in the study. However, the quantity of nitrate reductase (nap) in biofilm (877) was much lower than that in sludge (1933). This might be associated with the biodiversity of nitrate-reducing bacteria with different metabolism ways. In terms of nitrite reduction, a key gene (nirKS) was detected with similar abundances between the two samples showing further biochemical transformation from NO_2^--N to NO. Furthermore, nitrite reductase (nrf) able to catalyze DNRA[35] with relatively high abundances was noticed in our methanogenic system. A quite small number of assimilatory nitrate reductase and nitrogenase was also found, which participated in the nitrogen bioconversion. The identification of these functional genomes might constitute the complete nitrogen metabolism network.

Metabolic pathways

In this study, metagenomic analysis was used to reveal further metabolic pathways for understanding the transformation behavior of organics and nitrogen. In the organic metabolism given in Fig. 5, glycolysis was the first step, in which glucose was decomposed into pyruvic acid *via* Embden-Meyerhof pathway (EMP) by a series of genes. And then key genes participating both in the aerobic and anaerobic degradation of pyruvic acid were distinguished because of the microaerobic environment. In the aerobic fermentation of pyruvic acid, acetyl coenzyme A was directly formed and was then completely degraded into CO_2 through tricarboxylic acid (TCA) cycle. But in the anaerobic fermentation, pyruvic acid was firstly catalyzed into VFA (including acetic acid) by pyruvic acid lyase (pfl), and then into acetyl coenzyme A as a core precursor to methane production by genes (acs, ack and pta).[36,37] Under the action of gene cdh, acetyl coenzyme A was combined with 5,6,7,8-tetrahydromethanopterin (THM(S)PT) to form coenzyme M (CoM) and methyl-CoM under the role of gene mtr. Finally, methane could be yielded by reduction of methyl-CoM by key gene mcr. Besides, a few methyl-CoM might be reduced to Com-S-S-CoB and then back to CoM again by genes hdr and mvh, both of which continued to participate in the methane production pathway.[38] This metagenomic pathway information strongly indicated methane production from acetate oxidative decarboxylation in this study.

After the finish of methanogenesis, methane was at first oxidized into methanol by the action of methane oxygenase (mmo) under microaerophilic conditons.[39] We could convincingly believe that AMO was responsible for the CH_4 bioconversion. Afterwards, methanol was mainly further oxidized into formaldehyde and formate based on the detection of alcohol oxidase (MOX) and formaldehyde dehydrogenase (fdhA, gfa). Complete mineralization might also be included by gene (FDH and fds). A sequence of biochemical reaction steps indicates complete AMO-induced[40] enzyme systems in our study. We also tracked that a fraction of byproducts, *e.g.* acetic acid and methanol (respectively, from fermentation and AMO steps), were also utilized for possible denitrification.[41] Owing to these bioconversion pathways, glucose was thoroughly transformed into biogas (CH_4 and CO_2).

Figure 6 reveals nitrogen metabolic pathways. Distinctly, complete nitrate-nitrogen bioconversion routes are composed of nitrate reduction to nitrite, nitrite reduction to nitrogen oxide, nitrite reduction to ammonia, assimilatory nitrate (nitrite) reduction, and nitrogen fixation in our system. Even though the first step started at the NO_3^--N reduction to NO_2^--N, the involved genes were not the same. Dissimilatory nitrate reduction (denitrification and DNRA) was driven by genes nap and nar, while gene nas executed assimilatory nitrate reduction (ANR). Based on Fig. 4, the total abundances of nar and nap were always higher than nas, implying dissimilatory reduction pathways dominated in nitrogen removal. Subsequently, further nitrite reduction reactions included three different types of pathways, according to metagenomic analysis: 1 denitrification: NO_2^--N underwent a three-step reduction reaction into dinitrogen in the role of nirKS, nor, and nos, followed by NH_4^+-N formation through nitrogen fixation (NF) gene (nif); 2 DNRA: NO_2^--N could be directly reduced to NH_4^+-N with the participation of nirBD and nrf; 3 ANR: NO_2^--N assimilatory reduction to NH_4^+-N was executed by gene nirA. These also correctly explained that an extremely high nitrogen removal rate along with a fraction of ammonia remained in the final effluent in Fig. 2(a).

Implication of SMAMOD process

Figure 7 indicates a comprehensive metagenomic description of organics and nitrogen metabolism networks to explore the possible mechanism of SMAMOD in the USBBR system. In this graph,

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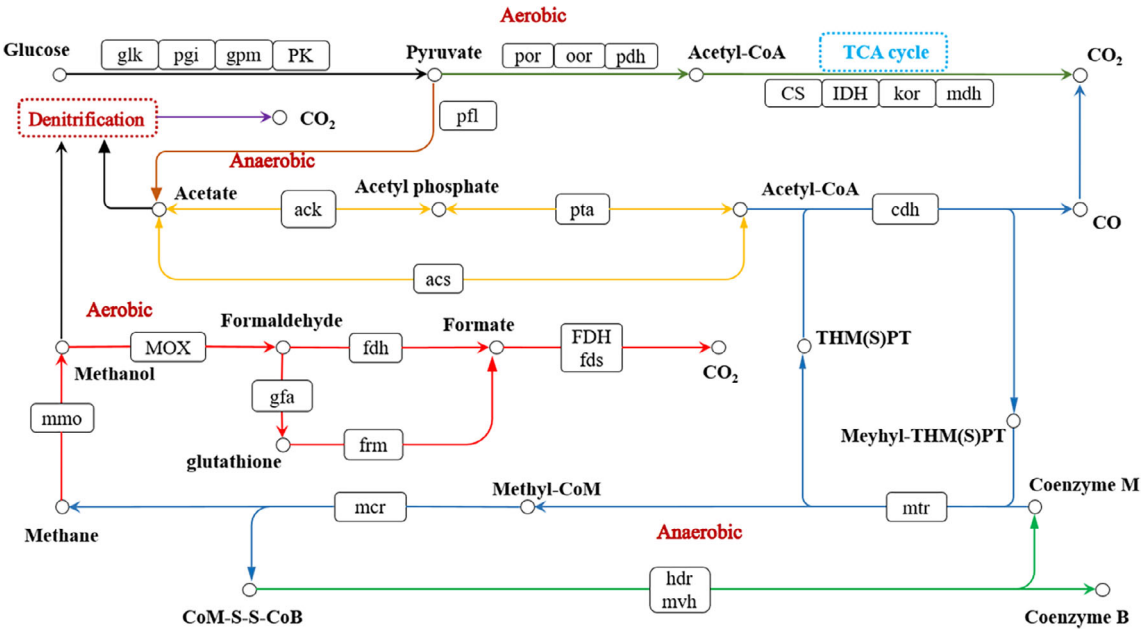


Figure 5. Genetic pathways for organic metabolism.

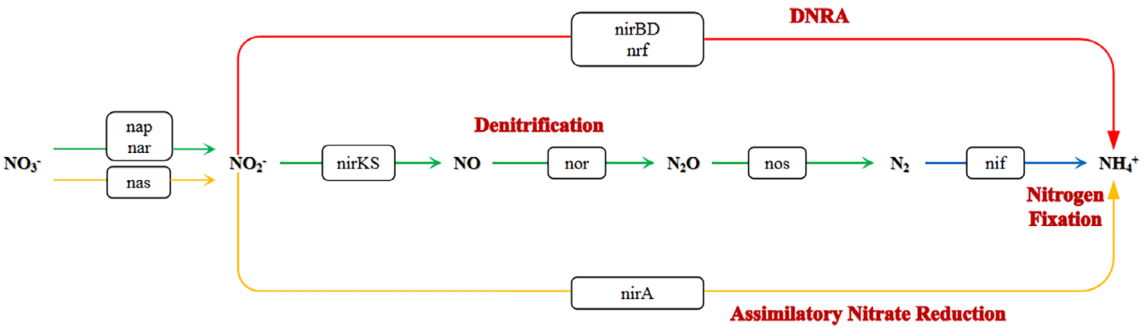


Figure 6. Genetic pathways for nitrogen metabolism.

left and right heatmaps represent the relative abundance of key functional bacteria and genes in the sludge and biofilm. As shown, glycolysis was the first step during glucose biotransformation. Masses of aerobic and anaerobic facultative bacteria (*Acinetobacter*, *Saccaribacteria*, *Longilinea*, *Levilinea*) as hydrolysis and acidification core genera were primarily found in the sludge bed. At the bottom, where glucose was rich, most of the glucose was able to be readily decomposed by these organic-degrading bacteria with highly rich glucose-fermentation enzyme systems. Determination of key functional genes in the sludge bed suggests anaerobic fermentation was a main pathway for further transformation of glycolysis products like pyruvic acid base on the heatmap, although its aerobic respiration *via* TCA cycle also existed in microaerobic hydrolysis process. Based on the anaerobic fermentation of pyruvic acid, acetyl coenzyme A, as an indispensable intermediate, participated in the formation of coenzyme M for methane-synthesis metabolic reaction through the action of key genes (mtr, mcr, hdr). As such, methane was successfully generated through decarboxylation of acetic acid, which was also confirmed by the definite identification of *Methanothrix* as a typical acetotrophic methanogen[27] in this study. Meanwhile, only a portion of acetic acids could be provided as a carbon source for denitrifiers due to their activity inhibition at significantly deceased pH levels.

In the biofilm phase, even if glucose was rather scarce, anaerobic micro-environments further sped up bioconversion of residual micromolecular intermediates such as organic acids (including acetate) into methane, when the formation of acetic acid was not a limiting step in producing methane. In this case, the similarly equal richness of methanation-associated microbes and metagenomes observed in both sludge and biofilm indicates methanogenesis was widely built up in the whole reactor. Further methane oxidation was also strongly evidenced by accurately probing of genus *Methylogaea* and genes (mmo and mox) with the potential AMO capacity based on metagenome results. Instead of homogeneous methanation, however, it was interesting to note that AMO was overwhelmingly developed in the biofilm. Why does AMO fail to occur in the sludge bed with seemingly stronger methanogenesis ability? We estimated that the setting of the packing bed as a beneficial habitat might prolong the contact time and enhance mass transfer in the bulk liquid between methane and biofilm, promoting effective *in situ* capture of methane by methanotrophic bacteria. Based on the abundance of key genes in AMO, methanol might be a main final byproduct, although the complete methane

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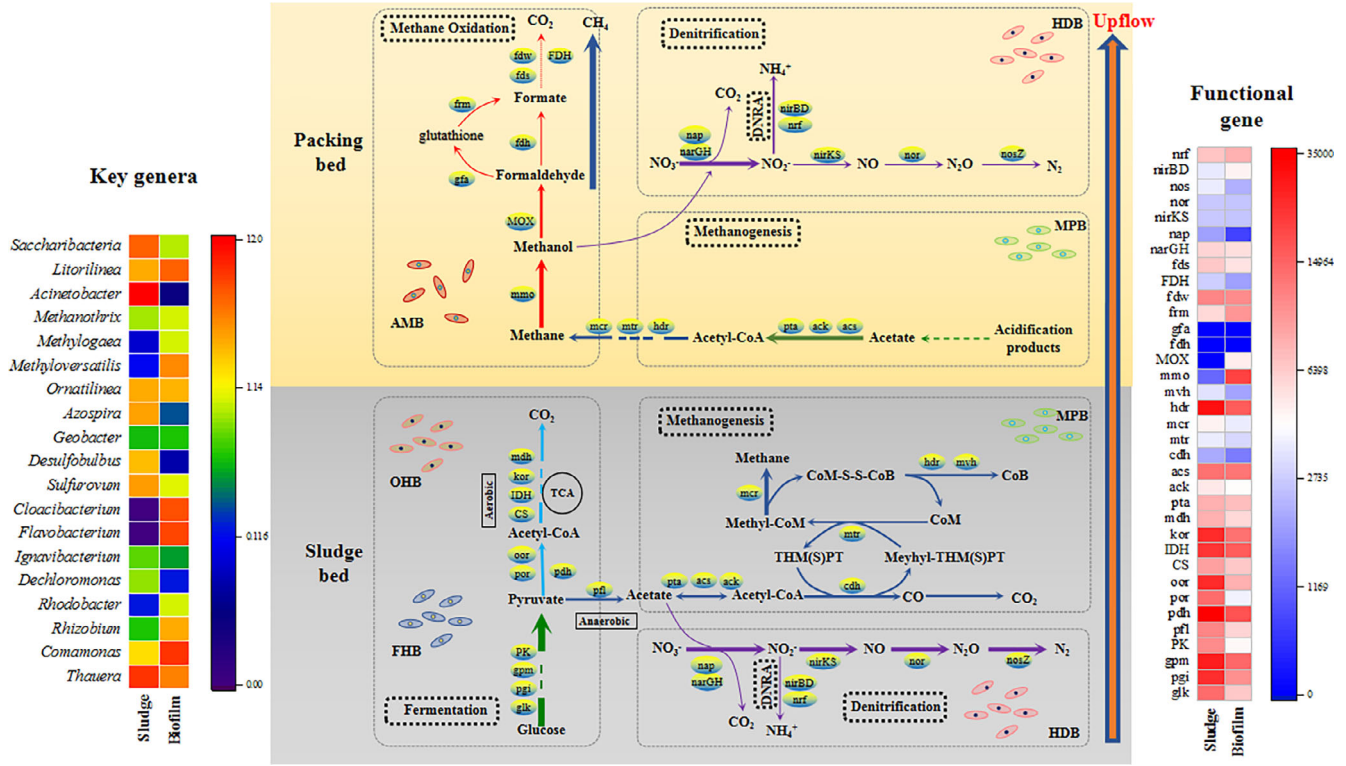


Figure 7. Possible mechanism of SMAMOD.

mineralization was possible with excessively low further methanol bioconversion genes (gfa, fdh). This methanol would be served as an alternative organic carbon for denitrification of nitrate not completely reduced in the sludge.

In the nitrogen biotransformation, dissimilatory reduction of nitrate was predominant, since a variety of facultative genera (*e.g. Thaura*, *Comamonas*, *Azospira*) conduced denitrification and even DNRA throughout the whole reactor. Determination of a series of important reductases (nar, nap, nirKS, nor and nos) revealed that complete denitrification step ($NO_3 \rightarrow NO_2 \rightarrow NO \rightarrow N_2O \rightarrow N_2$). Nevertheless, the type of denitrification was not identical along the vertical reactor, depending on different electron donors and functional denitrifiers. At the bottom of the reactor, fermentation products (*i.e.* acetic acids) were used by common denitrifiers, such as *Thauera*, *Comamonas*, *Rhizobium*, and *Azospira*, while only methanol acted as a carbon source by *Methyloversatilis* as alcohol-dependent denitrifiers at the reactor top. Owing to the vertical microbial interaction and substrates transfer, the combination of acetic- and methanol-denitrification caused efficient nitrogen removal in the present study. Some key nitrite reductase (nrf, nirBD) representing the existence of DNRA by some species of denitrifiers, especially *Flavobacterium*, *Cloacibacterium*, and *Geobacter*, could directly reduce NO_x-N to NH_4-N.

These metagenomics findings positively proved that the cooperative interaction of core genomes of methanogens, aerobic methanotrophs and denitrifiers allowed SMAMOD to entirely removing COD and nitrogen in the MUSBBR system. As for future application, one of the most remarkable potentials in SMAMOD is *in situ* availability of methane for easily cultivating aerobic methanotrophic microorganisms in a methanogenic reactor, hence enhancing nitrogen removal and relieving (dissolved) methane emission. Thus, it will offer a promising and robust one-reactor process for sustainable treatment of actual wastewaters such as chemical fertilizer, fish processing, feeding and landfill leachate wastewater. By applying SMAMOD, efficient C and N removal with partial biogas recovery and GHG reduction can be synergistically achieved. Yet, further SMAMOD researches on rapid enrichment of AMB, acquirement of high-purity methane using biogas separation techniques,[42] novel bioreactor exploitation for improving methane transfer and capture[43] as well as long-term optimization strategy will be still required.

CONCLUSION

In this study, a novel SMAMOD process achieving efficient organic and nitrogen removal as well as partial biogas recovery was developed in a single MUSBBR. The excellent performance was verified by microbial community and gene profiling analysis. Functional bacteria and functional genes were found to be involved in methanogenesis, aerobic methane oxidation, and denitrification. Genetic roadmap indicated complete pathways of aerobic/anaerobic fermentation, methane production, and methane aerobic oxidation in organic metabolisms, while denitrification, DNRA, assimilation nitrate reduction, and fixed nitrogen pathways were included in the nitrogen metabolism. Methanogenesis-, aerobic methane oxidation-, and denitrification-associated metagenomic cooperation and interaction led to SMAMOD.

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